REDCap homework 2026-27

Build the TUMAINI multi-centre study in REDCap

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At a glance

Deadline: Sunday 13 December 2026, 11:00 (Brussels time).

Work in groups of three. Each member is the data manager of one TUMAINI site.

Study: TUMAINI, a fictional cohort of adolescent girls at three sites in Tanzania (it replaces the PISTIC study of 2025). Read the document **TUMAINI protocol summary and CRFs** first.

Submit on Blackboard: the four files listed in section 9, one submission per group.

Server: redcap.uantwerpen.be. Purpose of the project: Practice / Just for fun. Use fake data only.

1 What you will build

A longitudinal REDCap project with **three arms**, **three sites** (Data Access Groups), **six instruments**, one instrument converted to a **survey**, **repeating** events and instruments, and the advanced features from the second lecture: branching logic, calculations, field embedding, piping, action tags and smart variables.

Part	Task	Points
A	Project, users and roles	5
B	Six instruments from the CRFs (five given, one you design)	25
C	Arms, events and form mapping	15
D	Repeating event and repeating instrument	10
E	Multi-centre set-up with Data Access Groups	15
F	Advanced features and the survey	20
G	Testing, production and the mobile app	10
	Total	**100**

2 Part A · Project, users and roles (5 points)

1. One group member creates a new project: **New Project**. Title: `TUMAINI 2026 - group number` (e.g. `TUMAINI 2026 - group 07`). Purpose: **Practice / Just for fun**. Start from an empty project.
2. In Project Setup, enable **Use longitudinal data collection with defined events** and **Use surveys in this project**.
3. User Rights: add the other group members, plus the course accounts `tafadzwa` and `redcap`.
4. Create two user roles: **Data Manager** (full rights, including Data Access Groups and User Rights) and **Site Staff** (data entry, no design, no user rights, export de-identified only). Put yourself and the two course accounts in Data Manager for now. In Part E you will move each group member to Site Staff.

3 Part B · Instruments (25 points)

Build the five forms in the CRF document, one REDCap instrument per form. Keep the item numbers in the field labels (e.g. `4. Is the participant 16 to 20 years old?`) so a monitor can compare the screen with the paper.

Instrument name	Form	Completed by	CRF pages
`selig`	Screening and eligibility	Study nurse	SELIG
`sege`	Screening genital examination	Clinician	SEGE
`fsx`	Sexual history and contraception	Participant (survey)	FSX
`fge`	Follow-up genital examination	Clinician	FGE
`mv`	Missed visit	Study nurse	MV
`loc`	Locator and knowledge check	Study nurse	you design it

Rules for every instrument

- Use short, lower-case variable names with a prefix per form (e.g. `sege_lmp`, `fge_lmp`). Never reuse a name.
- Code answers as on the CRF; use the same codes for the same answers everywhere (1 = Yes, 0 = No, 9 = Not applicable).
- Validate every date (`date_dmy`) and number (integer with a minimum and maximum). Mark identifiers (date of birth, phone, GPS).
- Turn every arrow on the CRF (→ If No, go to item 5) into **branching logic**. A skipped question must not be visible.

The eligibility logic on SELIG

- 1a is asked only if the participant is 18 or older; 2a only if she is younger than 18; 3a only if a parent or guardian consented.
- `eligible` is a calculated field: 1 if consent (or guardian consent plus assent) is given, items 4 to 8 are all Yes and items 9 to 11 are all No; otherwise 0.
- Show item 13 only if eligible. Show the arm (item 15) only if enrolled.

Your own instrument: LOC (locator and knowledge check)

Design it yourself. It must have four sections and use the features listed.

Section	Content	Must use
A · Locator	Phone numbers, how the participant may be contacted, GPS position of the home, a landmark	`@CHARLIMIT` on phone fields (Tanzanian numbers: the built-in phone validation only accepts North American numbers), `@LATITUDE` and `@LONGITUDE`, identifiers flagged, field embedding for the GPS pair
B · Knowledge	Which methods prevent pregnancy (checkboxes), with a follow-up question shown only when relevant	`@NONEOFTHEABOVE` for “I do not know”, `@HIDECHOICE`, branching such as `[kn_methods(2)] = '1'`
C · Logic check	A score from the knowledge answers, and a text version of it	a calculated field and `@CALCTEXT`
D · Follow-up	Is the participant eligible for follow-up? (calculated)	a smart variable: `[first-event-name][eligible]`

4 Part C · Arms, events and form mapping (15 points)

Define the three arms and their events exactly as below (Project Setup > Define My Events). Give each event the day offset and window shown. Then designate the instruments (Designate Instruments for My Events).

Arm 1 · Standard follow-up

Event	Unique event name	Day	Window	Instruments
Day 0	day_0_arm_1	0	-	selig, sege, fsx, loc
Month 3	month_3_arm_1	91	±14	fsx, fge
Month 6	month_6_arm_1	182	±14	fsx, fge
Month 9	month_9_arm_1	273	±14	fsx, fge
Month 12	month_12_arm_1	365	±14	fsx, fge
Unscheduled	unscheduled_arm_1	-	-	fsx, fge
Missed visits	missed_visits_arm_1	-	-	mv

Arm 2 · Intensive follow-up

Event	Unique event name	Day	Window	Instruments
Day 0	day_0_arm_2	0	-	selig, sege, fsx, loc
Month 1	month_1_arm_2	30	±7	fsx
Month 2	month_2_arm_2	61	±7	fsx
Month 3	month_3_arm_2	91	±7	fsx, fge
Month 4	month_4_arm_2	122	±7	fsx
Month 5	month_5_arm_2	152	±7	fsx
Month 6	month_6_arm_2	182	±7	fsx, fge
Month 9	month_9_arm_2	273	±14	fsx, fge
Month 12	month_12_arm_2	365	±14	fsx, fge
Unscheduled	unscheduled_arm_2	-	-	fsx, fge
Missed visits	missed_visits_arm_2	-	-	mv

Arm 3 · Reduced follow-up

Event	Unique event name	Day	Window	Instruments
Day 0	day_0_arm_3	0	-	selig, sege, fsx, loc
Month 6	month_6_arm_3	182	±14	fsx, fge
Month 12	month_12_arm_3	365	±14	fsx
Unscheduled	unscheduled_arm_3	-	-	fsx, fge
Missed visits	missed_visits_arm_3	-	-	mv

In the Reduced arm only the external genital examination is done. Hide the speculum and sample questions of FGE in arm 3 with the smart variable `[arm-number]`, e.g. `[fge_done] = '1' and [arm-number] <> '3'`.

5 Part D · Repeating (10 points)

- Make **Unscheduled** a repeating event in all three arms (a participant may come back more than once between visits).
- Make **mv** a repeating instrument in the **Missed visits** event: one instance per missed visit. Use a custom label that pipes the visit, e.g. `[mv_visit]`, so the instances are easy to tell apart.
- In your report, explain in two sentences the difference between a repeating event and a repeating instrument.

6 Part E · Multi-centre: Data Access Groups (15 points)

TUMAINI runs at three sites. Staff at one site must see only their own participants; the data managers see everything. REDCap does this with **Data Access Groups (DAGs)**: one project, one group per site.

1. Applications > **Data Access Groups**: create `Dar es Salaam`, `Mwanza` and `Mbeya`. Write down the unique group names REDCap gives them.
2. Each group member takes one site: assign yourself to that DAG and to the **Site Staff** role. Keep `tafadzwa` and `redcap` in the Data Manager role and **not** in a DAG, so they can see all sites.
3. Enable **record auto-numbering** (Project Setup > Additional customizations). Create records as a site user: REDCap now adds the DAG number in front of the record ID (e.g. `2-1`). Explain in your report why this is useful.
4. Enter **six fake participants per site** (18 in total): two in each arm, each with Day 0 complete and at least one follow-up visit, one unscheduled visit and one missed visit in the whole group.
5. Prove the isolation: take a screenshot of the Record Status Dashboard as a site user (only your site) and as a Data Manager (all sites).
6. Data Import Tool: download the import template, add three more fake participants for **your** site in a CSV with the column `redcap_data_access_group` (use the unique group name), and import them. Show the result in your report.
7. Make a report filtered by DAG and an export with the DAG column. Answer in your report: when would you choose separate projects per site instead of DAGs?

7 Part F · Advanced features and the survey (20 points)

Feature	Where	What to do
Field embedding	SEGE items 3 and 11; FGE items 2 and 6; LOC	Put the yes/no answers of the signs and the test results in a table inside a descriptive field (`{sign_red}`).
Piping	FSX 2c and 2e; MV; survey texts	`Of these [sex_n] times, how many times was a condom used?` Pipe the visit in the survey instructions with `[event-label]`.
Calculations	SELIG, FSX, LOC	Age from date of birth with `datediff()`; eligibility; times without a condom = times had sex minus times with a condom.
Validation check	FSX	A warning (descriptive field with branching) when condom uses are more than the times had sex.
Action tags	all forms	At least `@TODAY`, `@CHARLIMIT`, `@CALCTEXT`, `@HIDECHOICE`, `@NONEOFTHEABOVE`, `@READONLY`, `@HIDDEN-SURVEY`.
Smart variables	FGE, LOC, survey	`[arm-number]`, `[event-label]`, `[first-event-name]`.
Survey	FSX	Enable FSX as a survey: participants answer the sensitive questions themselves on a tablet. Write a welcome text with piping, open it at a visit with **Open survey** and complete it as a participant.

8 Part G · Test, production and the mobile app (10 points)

- Test every instrument with the fake records: each branch, each calculation, each validation. Fix what does not work.
- Check: every field has a label, the arms and events are mapped as in Part C, the calculations give the right numbers.
- Move the project to **Production** (Project Setup > Move project to production). Your request goes to the REDCap administrator.
- REDCap Mobile App: install it, set up the project with the code from Applications > Mobile App, enter one participant offline, then synchronise. Take a screenshot before and after synchronising.

9 What to hand in

File	How to get it
1. `TUMAINI_group_XX_project.xml`	Project Setup > Other functionality > **Download metadata only (XML)**. This is your whole project design in one file.
2. `TUMAINI_group_XX_DataDictionary.csv`	Designer > Data Dictionary > Download the current Data Dictionary.
3. `TUMAINI_group_XX_codebook.pdf`	Designer > Codebook > print to PDF.
4. `TUMAINI_group_XX_report.pdf`	At most 6 pages: who did what, screenshots for Parts E and G, and the short answers asked in Parts D and E.

Before you submit

□ The project is in Production, and `tafadzwa` and `redcap` are users with Data Manager rights and no DAG.

□ Three DAGs exist, each with at least nine fake participants, and each member is assigned to one DAG.

□ FSX is a survey and at least one response was entered as a participant.

□ No real names, phone numbers or locations anywhere. Fake data only.

□ Uploaded on Blackboard before Sunday 13 December 2026, 11:00 (Brussels time).

Every group member must build at least one full instrument and at least one advanced feature (branching logic, action tags, field embedding, repeating events, the multi-arm set-up or the Data Access Groups). Say who did what in the report.

Questions: ask them in the class or on the Blackboard discussion board, not by e-mail, so that everybody gets the same answer.